

Co-Design with Myself: A Brain-Computer Interface Design Tool that Predicts Live Emotion to Enhance Metacognitive Monitoring of Designers

Qi Yang

qy244@cornell.edu

Human Centered Design, Cornell University
Ithaca, New York, USA

Tianlin Zhao

Department of Computer Science, Cornell University
Ithaca, USA

Shuo Feng

College of Architecture, Art, and Planning, Cornell
University
Ithaca, USA
sf522@cornell.edu

Saleh Kalantari

Human Centered Design, Cornell University
New York, USA
sk3268@cornell.edu

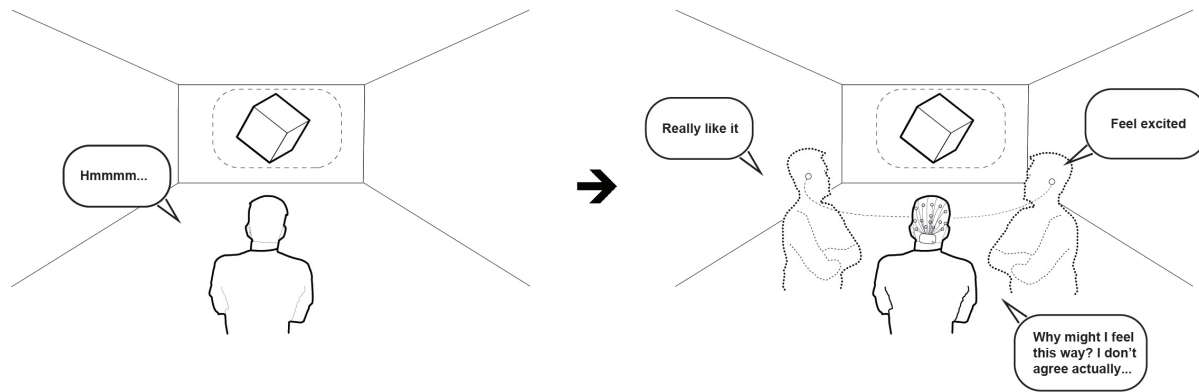


Figure 1: Concept Diagram of Co-design with Myself Using The “Multi-Self” Design Tool.

ABSTRACT

Modulating the feelings of confidence toward design options and metacognitive monitoring are important components of design intuition. We find not many creativity support tools explored the potential of biofeedback to aid design processes in this regard. In the current study, we present “Multi-Self,” a BCI-VR design tool that aims to enhance metacognitive monitoring during architectural design. It evaluates designers’ own valence and arousal to their design work and presents designers with their biofeedback visually in real time. A pilot proof-of-concept study with 24 participants was conducted to evaluate the feasibility of the tool. Interview responses regarding the accuracy of the feedback were mixed. The majority of the participants found the feedback mechanism useful and the tool

elicited metacognitive feelings and stimulated explorations of the design space while modulating subjective feelings of uncertainty.

CCS CONCEPTS

• **Human-centered computing** → *Systems and tools for interaction design.*

KEYWORDS

Creativity Support Tool, Brain-Computer Interface, Affective Computing, Biofeedback, Metacognition

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1 INTRODUCTION

Design is regarded as a non-linear and iterative process of divergent and convergent thinking. During the divergent thinking processes,

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designers frame the design space and explore as many design options as possible, followed by design evaluation and synthesis in the convergent thinking process. The iterative decision-making processes are full of uncertainty, which makes it challenging for designers to evaluate their designs in the early stages. It is almost impossible to foresee if an early decision will be successful due to the complexity it can unfold, especially in wicked design problems. Expert designers rely on intuition to make confident judgments, but the lack of self-reflection and awareness in their responses to different design options make many designers second guess themselves [17, 34, 45]. When designers ask themselves “Am I on the right track” or “If I have found the right design”, the answers could emerge from their intuitive confidence rather than rational evaluations. Sometimes, designers already have the answer but they hesitate to move forward. If lacking confidence, excessive hesitation could discourage designers from thoroughly exploring the design space and limit their creativity. Strong intuition and being able to maintain awareness of their self-confidence in early design stages are important for creative design practices [39].

Intuition, self-awareness, and feelings of confidence are entangled constructs in the design context. Despite the complex nature of intuition, it is described as a combination of unconscious evaluative processes and conscious reflective acts in the design practice [17]. The conscious reflective practice, also known as self-awareness or metacognitive monitoring, describes the process of “thinking about thinking” and consciously adopting effective mental strategies [1]. Indeed, being able to identify the strength and weaknesses of one’s own ideas and be aware of the rationales behind their decision-making will improve better idea evaluation and selection [43]. Feelings of confidence about design decisions, named epistemic uncertainty by Ball and Christensen, is also closely related to design evaluation [8]. According to this outlook, fluctuations in epistemic uncertainty may be an important trigger that prompts designers to alter or reinforce their mental approach to a design challenge. Too much uncertainty may limit the creative process by creating insecurity, apprehension, and creative paralysis [3]. Too little uncertainty during the early stages of design may contribute to design fixation. Some epistemic uncertainty can lead to deliberate counterarguments, initiate creative reconsideration, and potentially push a design forward [8, 41]. In theory, metacognitive monitoring plays a crucial role in modulating and keeping epistemic uncertainty within effective parameters, thereby avoiding the extremes of design fixation or creative paralysis [6, 10, 34]. As demonstrated, self-awareness and the ability to perceive accurate uncertainty and modulate a feeling of confidence in design are critical parts of strong design intuition. Building up strong intuition requires years of strategic practice. Even so, maintaining an accurate level of uncertainty is still challenging because internal mental states are difficult to perceive.

Recognizing the importance of intuitive feelings towards design, our motivation is to explore the possibility of a design tool that could modulate designers’ feelings of confidence and aid designers’ intuitive judgment through affective computing and biofeedback, specifically in an architectural design context. Affect is closely related to intuition, metacognition, and uncertainty. One common model of affect theory describes emotions that emerge in response to environmental stimuli as a blend of three independent factors:

(1) pleasure, also known as valence, which is an overall feelings of “positivity” or “negativity” towards something; (2) arousal, which describes the degree of “excitement”; and (3) dominance, which refers to feeling “unrestricted” or “restricted” in an environment [35]. Affects such as “gut feelings” are considered to be foundations of intuition [15]. Previous research has shown that metacognitive experiences are generally associated with improved affect, especially with feelings of pleasure [13, 14]. Uncertainty is usually associated with negative affect, but it can be positive when present in limited amounts in creative contexts [2]. Knowing the close relationship between affect and intuition, feelings of confidence, and metacognition, we envision a design tool that predicts if the user feels excited or positive towards the current design. More importantly, the tool will visualize those predictions to users in real time to help them perceive their internal states in a third-person perspective, in hopes of increasing self-awareness, modulating feelings of confidence, and improving intuition (Figure 1).

Prior work has used a variety of means to identify affect. One approach is grounded in evaluating facial expressions and gestures, which can be done computationally via emotion-recognition algorithms. Such evaluations, however, are based on secondary expressions of affect, which can vary quite a bit in different cultural, social, and individual contexts. Most importantly for our purposes, design is often a solitary process and when working alone many designers may not register their feelings via facial expressions. Another method that has been used to identify some forms of affect is electrodermal activity (EDA), but this physiological reaction is lacking in nuance and does not change fast enough to capture the rapidly changing responses that may occur during the design process. Using EEG measurements provides a much greater level of detail and directly evaluates brain responses. Previous studies have shown that affective states can be measured via EEG, including affective responses to architectural environments [4, 9, 49, 55]. Additional studies have classified pleasure, arousal, and dominance in numerous contexts and activities, providing benchmarks of accuracy that can be used to evaluate our BCI performance [19, 24, 27, 33, 52, 56].

In sum, the current project provides several important contributions to human-computer interaction research and to the development of creativity-support tools:

- We present a closed-loop BCI design tool that identifies and visualizes users’ real-time affect levels based on their EEG data.
- We have examined the BCI tool’s feasibility and usability by conducting user testing.
- The research evaluates the potential for real-time affect awareness to modulate feelings of confidence, stimulate metacognitive reactions, and support creativity for designers.
- We discuss the potential applications of this work and future avenues for affective computing and biofeedback in supporting creative design processes.

2 RELATED WORKS

Here we reviewed related works on EEG-based emotion detection, devices or tools that provide biofeedback to users, and human-AI co-creative tools.

2.1 Emotion Detection Based on BCI

Given its wide range of applications, automatic emotion classification is gaining the attention of scholars in HCI fields. Recent improvement in consumer-grade wearable EEG devices captures interest among the HCI community in using EEG for emotion classification [51]. Particularly focused on arousal and valence affective response, previous studies have used the EEG classification approach in the context of using music stimuli [38], music videos [5, 11, 48, 54, 57, 58], and video clips [29, 32, 40, 50, 53]. The immersive virtual reality (VR)'s ability to evoke emotion gives VR more interest to be used as a tool in emotion detection in general [16, 20, 22, 23, 28, 60]. In fact, the self-reported intensity of emotion was found to be significantly greater in immersive VR compared to similar content in non-immersive virtual environments [7]. In a relevant VR study, Marín-Morales et al. [33] used an EEG classification approach and reached an accuracy of 75.00% along the arousal dimension and 71.21% along the valence dimension (chance level: 58%) in the classification of four alternative virtual rooms. The same study was one of the first studies that developed an emotion recognition system using a set of immersive VR as a stimulus elicitation and discussed the application of the approach in architecture. The current study takes a further step and aims at developing a design tool based on emotion detection.

2.2 Biofeedback to Myself

Several studies have been done to help users be aware of their inner states through different modalities of feedback. One recent study developed glasses that recognized the wearer's emotions and helped to improve self-awareness. Another study developed Self-Interface, an on-body device attached to the spine that helped users perceive physiological signals by translating them to haptic feedback [21]. Similarly, another study helped users to perceive pupil dilation which is associated with different cognitive tasks by making pupil dilation audible. Their findings suggested most users were able to associate cognitive activities with the sounds, which elicited metacognitive awareness [12]. Besides haptic and audial feedback, a third study translated heart rate into shape-changing displays, which brought awareness of users' physiological states through visual feedback [59]. The Inner Garden project created a garden where mental states identified by EEG and breath sensors were translated into weather and day duration of the world. The device was meant to encourage self-reflection and foster mindfulness [44]. The majority of the studies focused more on developing the feedback system and less on user experiences in the context of an application. Nonetheless, those studies offer creative ideas about how biofeedback could be perceivable and suggest how perceiving biofeedback could benefit metacognitive awareness. Other applications of biofeedback have been explored such as modulating breathing [42], enhancing storytelling, promoting communication [46], [18] and gaming [25, 37]. One interesting study in the sleep facilitation context discussed design strategies that we found adaptable to our study in the design context [47]. The study pointed out the importance of motivating exploration, promoting responsiveness of BCI, and facilitating self-expression. Our study shared the same goal to increase user awareness in a different context. Though

we found few projects have specifically explored biofeedback's potential in decision-aiding and support intuition during the design process.

2.3 Co-Evaluate with an Agent

In a recent review article, Hsing-Chi Hwang [26] argued that AI-based co-creative tools can be divided into four types: "Editors" that enable specific design execution, "Transformers" that change design content to different forms, "Blenders" that mix creative elements together, and "Generators" that produce entirely novel creative products. We find it notable that this categorization schema does not include room for the type of stabilizing and biofeedback mechanisms that we envisioned for the current project. Thus, we propose a fifth type of co-creative tool, "Reflectors." They don't create, but instead, focus on empowering designers' self-awareness and enhancing their self-awareness and intuition during the process of addressing a design problem. Indeed, human-AI co-creative process will enrich the design space and dynamics of design iteration and we think it is equally important to develop a co-evaluating agent that helps users select their creations.

3 METHOD

We present the BCI design tool "Multi-Self" which aids architectural design processes. It was developed with two primary parts. The first was an application that allowed designers to create an interior lobby space in virtual reality. We chose the task of designing a lobby because it was a very common design problem and because a lobby's function and range of potential forms were quite flexible compared to other architectural features. We decomposed the process of lobby design into four components: the *Envelope* that determined the overall geometry and windows of the room, the *Layout* of furniture arrangements, the *Fixtures* that in this case were limited to lighting, and the *Color* palette for the various room components. In each of these categories, we pre-generated 14 to 31 models, which could be freely combined yielding 35,726,880 potential solutions in the design space.

To minimize motor actions and acquire clean EEG data of affect during the design selection, we developed a mixed-modality control function for this lobby design process, which combined head pointing and key presses. Users were able to focus on a design option through eye movements and then press buttons to adjust and confirm the selection. In addition to providing greater ease/fluidity in the design process, this approach helped to reduce large motion artifacts that can limit the effectiveness of the EEG classification.

The second primary component of Multi-Self was a BCI algorithm that categorized users' affective responses based on their EEG data (described in more detail below), and provided real-time biofeedback about these responses within the VR display. The form of this feedback was based on the emotion coordinate system developed by Lang [31]. The first quadrant displayed high-arousal and high-valence states (emotions generally described as "joyful" or "excited"). The second quadrant displayed high-arousal and low-valence states ("fearful" or "enraged"). The third quadrant displayed low-arousal and low-valence states ("boring" or "sad"), and the fourth quadrant displayed low-arousal and high-valence states ("relaxed" or "calm"). During the design process, users' live emotional

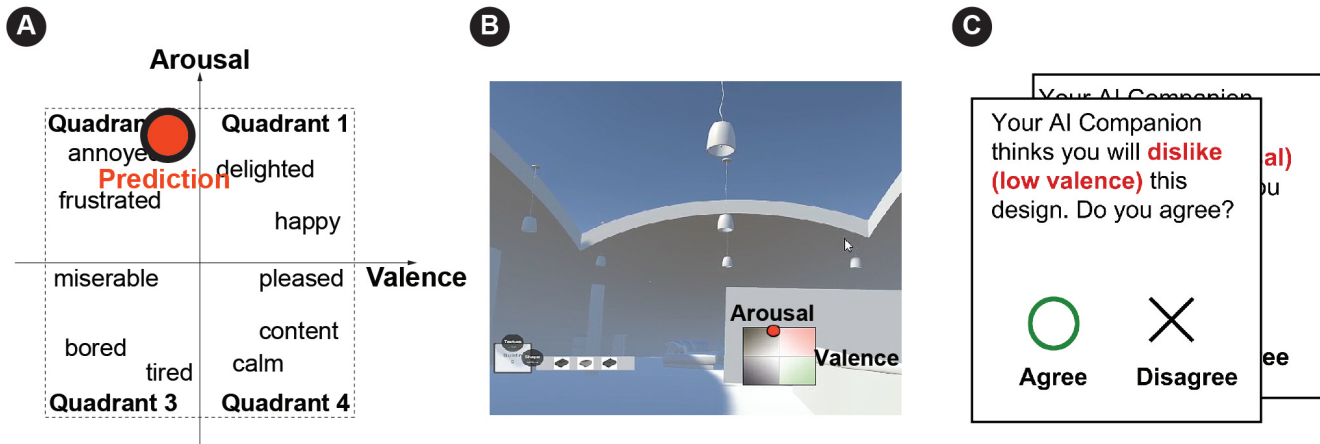


Figure 2: (A) Using the Emotion Coordinate System to Display the Results of EEG Classification; (B) Real-Time Visual Feedback in VR from Multi-Self; (C) Neurological Feedback and Pop-up Panels from "Multi-Self" in VR during the Validation Session.

feedback was visualized in the corner of the VR display as a moving colored point on this coordinate graph (Figure 2).

3.1 Participants

The pilot test was intended to provide preliminary feasibility and user-feedback data, rather than robust confirmation of scientific hypotheses. 24 participants with normal or corrected-to-normal vision were recruited through convenience sampling (word-of-mouth and announcement on departmental e-mail lists). Ten participants had an architectural design background. Their ages ranged from 19 to 36 ($M=22.9$, $SD=4.86$). 14 participants reported as female and 10 participants reported as male. Three participants indicated that they had previously used BCIs on rare occasions, and 21 participants indicated that they had never used BCIs. Four participants reported having more than 8 years of design experience; Six participants reported having 3–8 years of experience; Six participants reported having 1–3 years of experience; Eight participants reported having no design experience. We categorized designers with over 3 years of design experience as expert designers and designers with less than 3 years of experience as novice designers, adding up to 10 expert designers and 14 novice designers. Six of the participants reported having less than six hours of sleep the previous night—a notable factor that could potentially affect neurological responses and mood states. The overall study protocol was approved by the Institutional Review Board prior to the start of research activities.

3.2 Experiment Procedure

Sessions were conducted for one participant at a time. Each participant was first asked to complete a pre-experiment survey. The researchers then assisted the participant with fitting the EEG and VR equipment (Figure 3). The participant was asked to engage in a training session so that the BCI could become familiar with their individual emotional responses. This involved observing a panoramic VR image of an architectural environment for 10 seconds while we recorded EEG data and then completing the brief SAM scale [31] to report arousal and valence. The process was repeated until 49

pre-selected environments were experienced, which took around 15 minutes in total.

Next, participants were placed into a practice session with limited design options to familiarize themselves with the head-pointing and Xbox adaptive controller controls. A research assistant gave instructions such as “please change the lobby envelope to a different model,” and the practice continued until the participant was able to fluidly use the controls. During this time, the researchers pre-processed the data collected during the training session and used it to complete the machine-learning classification. We trained two 3-class, user-specific, linear Support Vector Machine (SVM) models to predict high, neutral, and low arousal, and high, neutral, or low valence. This process took around 5 to 10 minutes, and we explained to the participants that we were analyzing the data while they learned the design controls.

Each participant then continued to the testing/validation session. In this part of the experiment, all of the lobby design options were available to the participant, and they were asked to experiment with different options to create an effective design. We used two techniques to validate the accuracy of the feedback tool. First, when the participant made any change to the design, they were prompted to wait for 3 seconds, during which their brain activities were analyzed and classified by the BCI tool. The participants were then shown the neurological feedback and asked through a pop-up panel if they believed it to be accurate. For example, if the model predicted the user was feeling high arousal, the popup window would say “Your AI companion thinks you would feel excited about this design change, do you agree? (Yes/No).” After making another design change, the participant would experience the second validation method, which asked them to report arousal and valence via the SAM scale without seeing the neurological feedback. These two validation methods alternated until they had each occurred 6 times.

Finally, the participants were allowed to continue freely working on lobby designs until they were satisfied with the outcome, during which time they could view the ongoing feedback from the



Figure 3: Experiment Setting of the Augmented Co-Design Process.

BCI response-tracking tool. Each participant was asked to complete three different lobby designs in this fashion, which generally took about 15 minutes. When satisfied with their work, the participants removed the VR and EEG equipment and completed a short debriefing interview about their experiences (Figure 4).

3.3 Environmental Stimuli Used for Training the BCI

The most common environmental datasets used for emotional response analysis in prior research, such as the DEAP dataset [30], were in the format of two-dimensional videos. We were able to find only two existing datasets offering immersive 3D videos for this purpose. One of them included outdoor human and animal activities and was thus unsuited for architectural design analysis [36]. Another small existing 3D dataset did include indoor environmental stimuli, but this amounted to only four samples, which was insufficient for our machine learning needs [33]. Therefore, we curated our own immersive 3D environmental stimuli to stimulate affective responses from the participants. Since we assumed participants' emotional responses to architectural environments would be highly personalized, we did not prescribe arousal and valence labels to these images by asking experts to rate each stimulus. Instead, we tried to create a selection that would be likely to elicit a broad range of emotional responses. In the first step of this process, we download one hundred 360-degree panoramic images from the internet. Some of them were images of real-world environments and others were renderings of imagined spaces. To remove stimuli that would be unlikely to provoke an affective response, two judges with over 5 years of architectural design experience were asked to view all of

the downloaded images via the same HTC-Vive headset used in the experiment and to report their arousal and valence levels related to each image using the SAM scale. We removed images that received "moderate" scores from both judges on each axis (scores of 2 or 4, out of a 1–5 range). The final selection of 50 images was made with the goal of obtaining a balance between "high" (5), "neutral" (3), and "low" (1) scores on each axis, to ensure that each quadrant on the Emotion Coordinate System was reasonably represented. Due to one operation error from the experimenter, the first stimulus was systematically not presented to all the participants. Eventually, participants experienced 49 stimuli.

3.4 Outcome Measures

We measured machine learning performance by three metrics: (1) offline validation accuracy, (2) online user agreement with the displayed neurological feedback, and (3) consistency between online self-reporting and unseen neurological feedback. The purpose of this pilot study was to conduct proof-of-concept testing and obtain initial user feedback. Because our training data was imbalanced, we measured the imbalance ratio calculated by the division between the number of the class with the most samples and the number of the class with the least samples.

Qualitative data were gathered during semi-structured exit interviews to understand the potential impacts of emotional feedback on metacognitive feelings. From the transcription, we rated users' subjective trust level on an ordinal scale ("low", "medium", and "high").

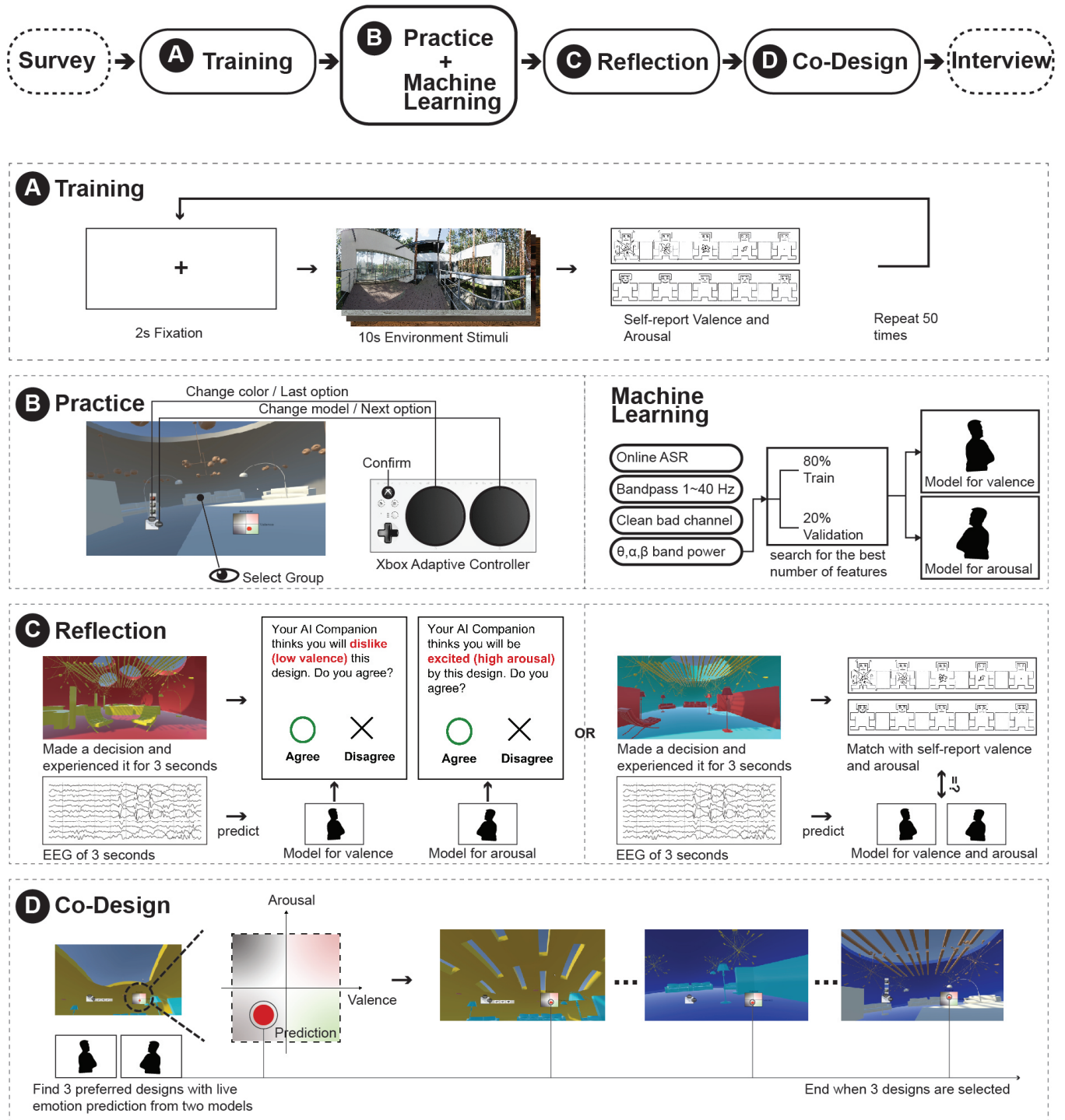


Figure 4: Schematic Overview of the Experiment Procedure: (A) Participants Experience Fifty Panoramic Environmental Stimuli and Self-Report Valence and Arousal; (B) Participants Practice Using Xbox Controller to Change Design Options; (C) Participants Express Agreement with "Multi-Self" and Report Valence and Arousal of the Selected Designs; (D) Participants Select Three Designs without Interruption with Real-time Emotion Predictions.

4 CONCLUSIONS AND DESIGN SUGGESTIONS

We found that participants expressed moderately strong trust in the predictions of our BCI design tool. Based on a content analysis of the interviews, the tool motivated metacognitive feelings, especially when the design tool's predictions mismatched participants' intuitions. Detailed results will be discussed in the following full paper.

In terms of suggesting designs for the future, we would like to offer some perspectives that we believe are worth considering. Firstly, it's important to recognize uncertain design behaviors and interactions, which can be used to modulate subjective uncertainty. Secondly, finding a balance between design engagement and reflective processes is crucial. Lastly, creating multi-modal feedback that doesn't interfere with the design process is key.

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